# IS333 Assignment 1
# Semester 1, 2025


Group 40:

S11220337 – George Fong

S11219345 – Akash Mishra

S11190968 – Perry Siola

S11197324 – Ravinesh Narayan

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# Part A: Activity Durations

The table below shows all the activity labels, the predecessors of the activities and the estimated Durations of each activity in the project.

*To determine the estimated duration of each activity, an appropriate time estimation technique was applied in this part of question.*

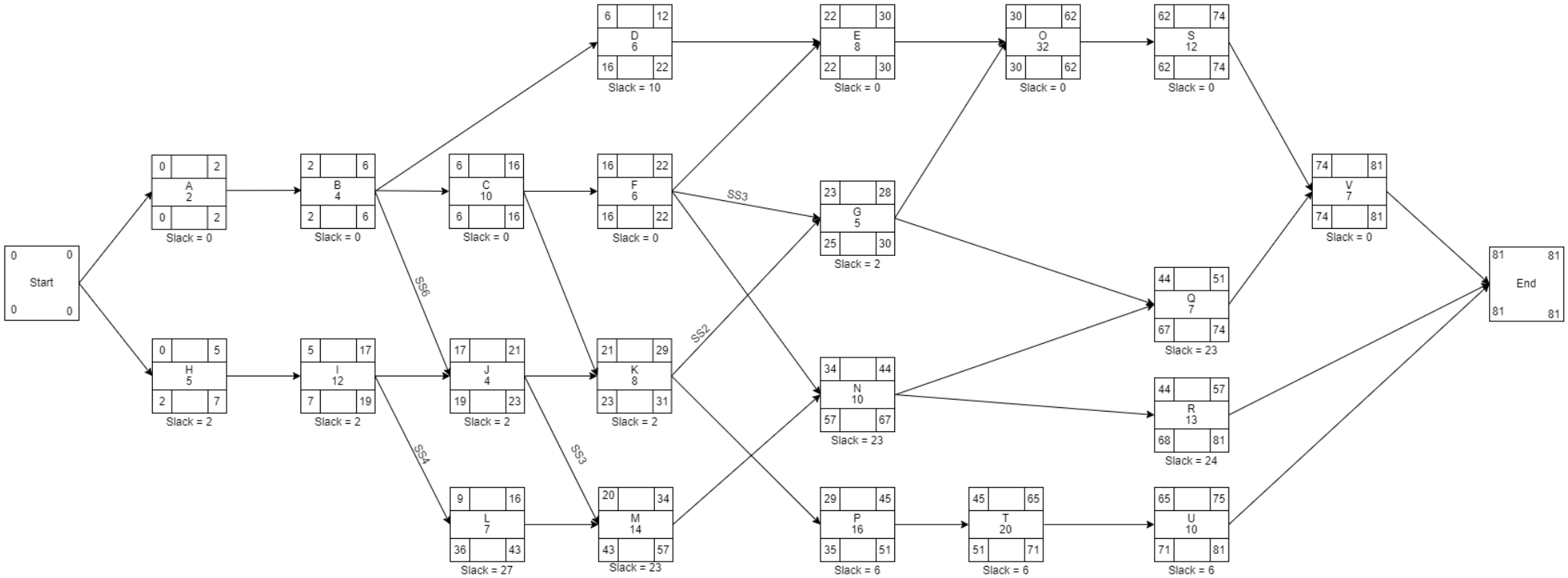
| Activity label | Optimistic | Most Likely | Pessimistic | Working<br>T = (a+4(m)+b) /6 | Duration (Days) |
|---|---|---|---|---|---|
| **A** | 1 | 2 | 5 | (5+4(2)+1) /6 | 2.333333 |
| **B** | 3 | 4 | 5 | (5+4(4)+3) /6 | 4 |
| **C** | 8 | 10 | 12 | (8+4(10)+12) /6 | 10 |
| **D** | 2 | 6 | 7 | (7+4(6)+2) /6 | 5.5 |
| **E** | 6 | 8 | 10 | (6+4(8)+10) /6 | 8 |
| **F** | 5 | 6 | 7 | (5+4(6)+7) /6 | 6 |
| **G** | 3 | 4 | 8 | (3+4(4)+8) /6 | 4.5 |
| **H** | 4 | 5 | 7 | (4+4(5)+7) /6 | 5.166667 |
| **I** | 10 | 12 | 14 | (10+4(12)+14) /6 | 12 |
| **J** | 3 | 4 | 7 | (3+4(4)+7) /6 | 4.333333 |
| **K** | 7 | 8 | 11 | (7+4(8)+11) /6 | 8.333333 |
| **L** | 5 | 7 | 9 | (5+4(7)+9) /6 | 7 |
| **M** | 10 | 14 | 16 | (10+4(14)+16) /6 | 13.666667 |
| **N** | 9 | 10 | 12 | (9+4(10)+12) /6 | 10.166667 |
| **O** | 27 | 32 | 37 | (27+4(32)+37) /6 | 32 |
| **P** | 14 | 16 | 18 | (14+4(16)+18) /6 | 16 |
| **Q** | 5 | 7 | 9 | (5+4(7)+9) /6 | 7 |
| **R** | 10 | 13 | 16 | (10+4(13)+16) /6 | 13 |
| **S** | 11 | 12 | 14 | (11+4(12)+14) /6 | 12.166667 |
| **T** | 18 | 20 | 22 | (18+4(20)+22) /6 | 20 |
| **U** | 9 | 10 | 12 | (9+4(10)+12) /6 | 10.166667 |
| **V** | 6 | 7 | 10 | (6+4(7)+10) /6 | 7.333333 |

## Part B: Activity Table

The table below presents the predecessors for all activities, along with their Start-to-Start (SS) relationships and the respective durations of each activity.

| Activity label | Predecessors | Duration (Days) |
|---|---|---|
| **A** | - | 2 |
| **B** | A | 4 |
| **C** | B | 10 |
| **D** | B | 6 |
| **E** | D, F | 8 |
| **F** | C | 6 |
| **G** | F(SS3), K(SS2) | 5 |
| **H** | - | 5 |
| **I** | H | 12 |
| **J** | B(SS6), I | 4 |
| **K** | C, J | 8 |
| **L** | I(SS4) | 7 |
| **M** | J(SS3), L | 14 |
| **N** | F, M | 10 |
| **O** | E, G | 32 |
| **P** | K | 16 |
| **Q** | G, N | 7 |
| **R** | N | 13 |
| **S** | O | 12 |
| **T** | P | 20 |
| **U** | T | 10 |
| **V** | Q, S | 7 |

# Part C: AON Diagram



## Part D: Project Duration

The project will take **81 days**. The completion date is 1st August 2025 (Thu 01/08/25) derived from MS Project after taking all non-working days into account.

## Part E: Distinct Paths

1. A-B-D-E-O-S-V = 71 days
2. A-B-C-F-E-O-S-V = 81 days
3. A-B-C-F-G-O-S-V = 75 days
4. A-B-C-F-G-Q-V = 38 days
5. A-B-C-F-N-Q-V = 46 days
6. A-B-C-F-N-R = 45 days
7. A-B-C-K-G-O-S-V = 74 days
8. A-B-C-K-G -Q-V = 37 days
9. A-B-C-K-P-T-U = 70 days
10. A-B-J-K-G-O-S-V = 70 days
11. A-B-J-K-G-Q-V = 33 days
12. A-B-J-K-P-T-U = 66 days
13. A-B-J-M-N-Q-V = 49 days
14. A-B-J-M-N-R = 48 days
15. H-I-J-K-G-O-S-V = 79 days
16. H-I-J-K-G-Q-V = 42 days
17. H-I-J-K-P-T-U = 75 days
18. H-I-J-M-N-Q-V = 58 days
19. H-I-J-M-N-R = 57 days
20. H-I-L-M-N-Q-V = 54 days
21. H-I-L-M-N-R = 53 days

## Part F: Critical Paths

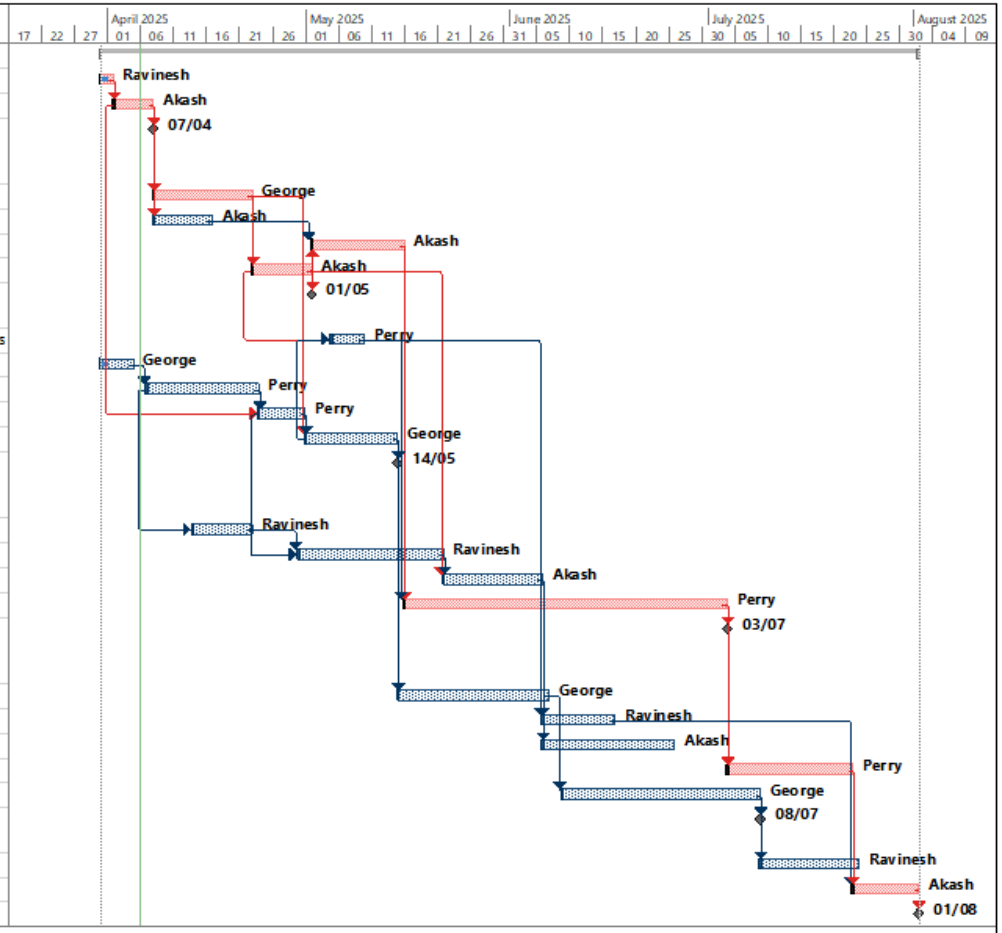
Critical Path: A-B-C-F-E-O-S-V = 81

## Part G: Slack

D = 10 days, G = 2 days, H = 2 days, I = 2 days, J = 2 days, K = 2 days, L = 27 days, M = 23 days, N = 23 days, P = 6 days, Q = 23 days, R = 24, T = 6 days, U = 6 days.

# Part H: Gantt Chart

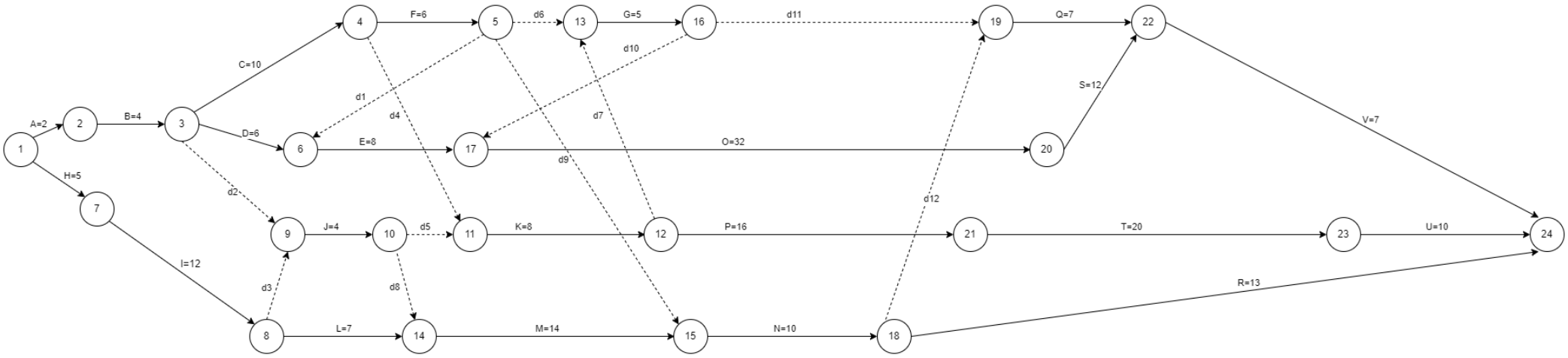
| ID | Task Name | Duration | Start | Finish | Predecessors |
|---|---|---|---|---|---|
| 0 | **Project** | **81 days** | **Mon 31/03/** | **Fri 01/08/2!** | |
| 1 | A | 2 days | Mon 31/03/2 | Tue 01/04/25 | |
| 2 | B | 4 days | Wed 02/04/2 | Mon 07/04/2 | 1 |
| 3 | Casino Infrastructu ready | 0 days | Mon 07/04/25 | Mon 07/04/25 | 2 |
| 4 | C | 10 days | Tue 08/04/25 | Tue 22/04/25 | 2 |
| 5 | D | 6 days | Tue 08/04/25 | Wed 16/04/2 | 2 |
| 6 | E | 8 days | Fri 02/05/25 | Thu 15/05/25 | 5,7 |
| 7 | F | 6 days | Wed 23/04/2 | Thu 01/05/25 | 4 |
| 8 | Casino floor fully | 0 days | Thu 01/05/25 | Thu 01/05/25 | 7 |
| 9 | G | 5 days | Mon 05/05/2 | Fri 09/05/25 | 7SS+3 days,13SS+2 days |
| 10 | H | 5 days | Mon 31/03/2 | Fri 04/04/25 | |
| 11 | I | 12 days | Mon 07/04/2 | Wed 23/04/2 | 10 |
| 12 | J | 4 days | Thu 24/04/25 | Wed 30/04/2 | 2SS+6 days,11 |
| 13 | K | 8 days | Thu 01/05/25 | Wed 14/05/2 | 4,12 |
| 14 | Security & Surveillance Operational | 0 days | Wed 14/05/25 | Wed 14/05/25 | 13 |
| 15 | L | 7 days | Mon 14/04/2 | Tue 22/04/25 | 11SS+4 days |
| 16 | M | 14 days | Wed 30/04/2 | Wed 21/05/2 | 12SS+3 days,15 |
| 17 | N | 10 days | Thu 22/05/25 | Thu 05/06/25 | 7,16 |
| 18 | O | 32 days | Fri 16/05/25 | Thu 03/07/25 | 6,9 |
| 19 | Regulatory Approval Secured | 0 days | Thu 03/07/25 | Thu 03/07/25 | 18 |
| 20 | P | 16 days | Thu 15/05/25 | Fri 06/06/25 | 13 |
| 21 | Q | 7 days | Fri 06/06/25 | Mon 16/06/2 | 9,17 |
| 22 | R | 13 days | Fri 06/06/25 | Wed 25/06/2 | 17 |
| 23 | S | 12 days | Fri 04/07/25 | Tue 22/07/25 | 18 |
| 24 | T | 20 days | Mon 09/06/2 | Tue 08/07/25 | 20 |
| 25 | Soft Opening & | 0 days | Tue 08/07/25 | Tue 08/07/25 | 24 |
| 26 | U | 10 days | Wed 09/07/2 | Wed 23/07/2 | 24 |
| 27 | V | 7 days | Wed 23/07/2 | Fri 01/08/25 | 21,23 |
| 28 | Grand Open | 0 days | Fri 01/08/25 | Fri 01/08/25 | 27 |




Project: Project
Date: Sun 06/04/25

Task | Split | Milestone | Summary | Project Summary | Inactive Task | Inactive Milestone | Inactive Summary | Manual Task | Duration-only | Manual Summary Rollup | Manual Summary | Start-only | Finish-only | External Tasks | External Milestone | Deadline | Critical | Critical Split | Progress | Manual Progress

# Part I: AOA Diagram with FS Only



# Part J: Resource Conflict and Techniques To Apply

## 1. Activity B (4) and H (5) running in parallel.

Resource Conflict:
Early tasks in the same project are overlapping and could be using the same resource, leading to over resource allocations and possible delays.
Solution:
Apply the critical chain project management methodology by buffering accordingly. Add a feeding buffer between B and H. Prioritize the activity that falls on the critical path and push the non-critical tasks to avoid multitasking (Pinto, 2019).

## 2. Activity C (10), D (6) and I (12) running in parallel.

Resource Conflict:
Assume the same person is required for A, D and I resulting in an overloaded resource. For example, Bob is needed for C, D and I, considering that the normal workday is 8 hours, he cannot be working 24 hours a day.
Solution:
Apply the critical chain project management methodology, first apply resource leveling. Pinto (2019) suggests working backwards from the end of the project to eliminate sources of conflict and selecting the option that minimizes total network duration.

## 3. Activity F (6), J (4) and L (7) running in parallel.

Resource Conflict:
Assume the same resources are required for F, J and L resulting in an overloaded resource.
Solution:
Use resource loading. As Pinto (2019) states we use resource loading to identify and combat situations like this by assigning appropriate resources to the appropriate degree for the network activities.

## 4. Activity P (16) and R (13) running in parallel.

Resource Conflict:
Heavy tasks running in parallel could lead to a burnout or decreased performance in a situation where a person is working on both these tasks, even at different time periods.
Solution:
Use crashing to reduce the duration of one or both of these tasks by allocating additional resources (Pinto 2019). This shortens the overall schedule and gives the overloaded person more time to recover and reduces the risk of burnout.

## 5. Activity O (32) and T (20) running in parallel.

Resource Conflict:
Tasks with very long durations could be overburdening the same resources.
Solution:
Apply critical chain resequencing, identify which task is on the critical chain and reschedule the non- critical one, add buffers to minimize risk and then apply resource levelling and loading (Pinto 2019).

# References

Alexander, M. (2024, March 28). How To Do Resource Allocation: Key Steps & Template. The Digital Project Manager. https://thedigitalprojectmanager.com/projects/managing-schedules/resource-allocation/

Pinto, Jeffrey. Project Management: Achieving Competitive Advantage, Global Edition, Pearson Education, Limited, 2019. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/uspfjebooks/detail.action?docID=5720015. Created from uspfj-ebooks on 2025-04-03 10:42:33.

https://drive.google.com/drive/folders/14vcWrSlds5LH0s58MyX8u0gf4eoDF5eS

# Appendix

## 1.1 TEAM OF REFERENCE

### *1.2 GROUP MEMBER DETAILS*

The assignment group consists of four 3rd year Software engineering students:

- Ravinesh Narayan is equipped with excellent analytical skills and has an interest in coding for programs. He can implement peer-reviewed strategies by actively engaging in group assessments and taking the lead as the project manager.
- Akash Mishra is a vibrant individual who possesses strong communication skills. He understands that project management is vital and has taken on the role of compiling the project documentation very carefully. The peer-reviewed strategies that Akash puts in include the good distribution of work among team members and timeliness in the project is also maintained. His organizational skills keep the project on track.
- George Fong believes that the domain of software engineering is challenging and builds personal resilience. He can develop diagrammatic representations of the overview of the project. The peer-reviewed strategies that George possesses include analyzing network activities and documenting them in detail. His diagrams contribute much to the team's understanding of the casino project.
- Perry Siola is a creative who has an interest in Draw.io and MS Word. He is developing his skills in the creation of professional diagrams. His peer reviewed strategies include researching resource conflicts and their solutions. He contributes to the project by putting his effort into ensuring that the diagrams being used are visually appealing.

### *1.3 SKILLS INVENTORY*

| Team Members | Ravinesh Narayan | Akash Mishra | George Fong | Perry Siola |
|---|---|---|---|---|
| **Technical Skills** | Good with Draw.io, Word, Visio, MSProject. | Good with Draw.io, Word, Visio, MSProject. | Good with Draw.io, Word, Visio, MSProject. | Good with Draw.io, Word, Visio, MSProject. |
| **Analysis Skills** | Good analysis skills. | Good analysis | Good analysis | Good with |

| | | | | |
|---|---|---|---|---|
| | | skill | skills | analysis skills |
| **Designing Skills** | UML Designing knowledge | UML Designing knowledge | UML Design knowledge | UML Design knowledge |
| **Research Writing Skills** | Excellent research skills | Able to research and document | Basic Research skills | Good in academic writing and literature review |
| **Presentation Skills** | Excellent presentation skills. | Able to present and answer questions | Good oral communicatio n skills and fluent English | Good in creating PowerPoint presentations |
| **Managerial Skills** | Manage the compilation of documents and ensure cooperation of team members. | Work cooperatively and manage tasks assigned | Introductory level management skills | Good in code review processes |
| **Document Processing and Compilation Skills** | Excellent document processing and compilation skills | Able to analyze document | Able to compile document | Good in creating system documentatio n |
| **Programming and Development Skills** | Java, C++, SQL, HTML, CSS and Visual Basic | Java, SQL, HTML and C++ | C++, Java, SQL and HTML | Java, HTML, C++ and SQL |

### *1.4 RESOURCES USED*

- Microsoft Project
- Microsoft Visio
- Microsoft Word
- Draw.io
- Lucid Chart

### *1.5 TEAM ROLES*

| **Team Members** | **Roles** |
|---|---|
| Ravinesh Narayan | Project Manager |
| George Fong | Project Scheduler |
| Akash Mishra | Project Scheduler |
| Perry Siola | Project Scheduler |

### *1.6 PEER REVIEW STRATEGIES*

Each aspect has been reviewed by each member. Peer review strategies comprise each aspect being divided and assigned to a team member based on what they are good at and learn from it. Ravinesh lead the casino project analysis, Akash, George and Perry also did their own individual analysis. The first draft consisted of cross-critical analysis of each other's work.

Those ideas that were missed were later shared and encouraging and motivating each other in pairs led to phase two which was amendment. The third phase covered completion with proofreading and final peer review before compilation.

### *1.6.1 DECISION MAKING POLICY*

All decisions were made with the agreement of every team member. Considering the ideas and interests of all parties. Each decision followed a process involving thorough research, careful planning and rescheduling.

### *1.6.2 CODE OF CONDUCT AND ETHICAL ISSUES*

The Australia Computer Society Code of Ethics has been followed to the full. All team members kept an ethical approach while at the same time improving the project's quality. The ethic for professional development has been emphasized through motivation, learning and teaching and brainstorming.

### *1.6.3 TEAM CONTRIBUTION*

## *1.7 LEADERSHIP*

Leadership roles were rotated among team members to align with each person's strengths in different aspects of the project. This developed good trust within the team members to implement a shared workload.

## *1.8 COMMUNICATION & COLLABORATION*

Communication tools like Viber and face to face meetings were used as needed. Google Docs proved highly effective for collaboration, facilitating idea-sharing and brainstorming.

# IS333 Project Management Semester 1, 2025 Assignment 1 Mark Allocation Agreement

After having discussed as team, we recommend the following mark allocation to each team member based on contribution or lack of it throughout the assignment.

| Member ID | Percentage of assignment 1 marks | Task Completed |
|---|---|---|
| S11197324 | 100% | All |
| S11220337 | 100% | All |
| S11190968 | 100% | All |
| S11219345 | 100% | All |

Signed

| Member name | ID | Signature |
|---|---|---|
| Ravinesh Narayan | S11197324 | |
| George Fong | S11220337 | |
| Perry Siola | S11190968 | |
| Akash Mishra | S11219345 | |